

- There are 3 Questions for 12 marks.
- Write your answers neatly, legibly and logically. Keep your proofs thorough. Your proofs must follow from the basics discussed in class.
- You do not need any calculators.
- Good luck!

1. [2] What is the limit of the following sequences

(a) $(1 + 1/2n)^n$

(b) $(-1)^n n! / n^n$

2. [4] What is the radius of convergence and limit of the following series

(a) $1 - x + x^2 - x^3 + x^4 - \dots$

(b) $\sum_{-\infty}^0 2^i x^i$

(c) $\sum_{-\infty}^0 a_i x^i$ where $a_i = \begin{cases} 2^i & \text{for } i < 0 \\ (-1)^i & \text{otherwise} \end{cases}$

3. (a) [2] Using the series expansion of $1/(1+x)$, derive the Taylor series for $\ln(1+x)$ about $x=0$.

(b) [2] What is the radius of convergence of this series ?

(c) [2] Give an approximation for $\ln(1.5)$ up to the second decimal place (you may use, if required:
 $1/64 \approx 0.015, 1/160 \approx 0.006$).